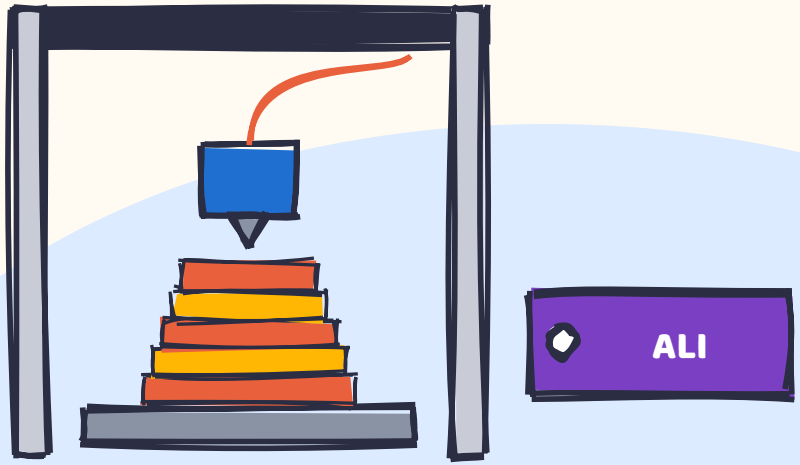
Junior Makers · Class 10

3D-Designed Name Badge & Sensor Mount



Design it on a computer. Print it. Use it on your robot!

Tinkercad 3D | Measuring in mm | Design to fit

Did you know?
Doctors 3D print custom hands and arms for children. Each one is designed to fit just one child.

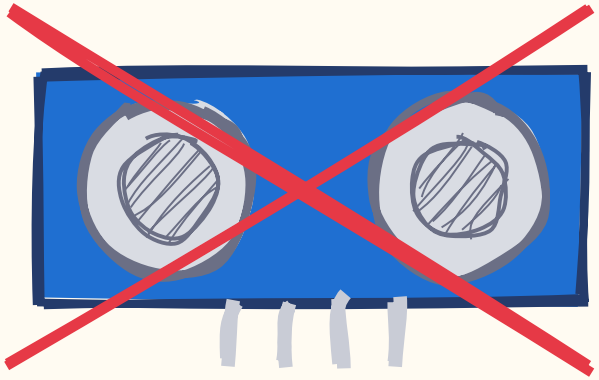
LEARN

Let's think...

How do engineers make something new?

The robot needs a part

Our robot's distance sensor is stuck on with tape. It wobbles and points the wrong way. No shop sells a holder for **our** robot. What can we do?



HC-SR04
wobbly tape!

Buy one? **Glue it?** **Design it!**

Today: we design our own parts on a computer, starting with a **name badge**, then a **sensor mount** for our robot.

Learning objectives

By the end of this class I can…

1	**Layers:** explain how a 3D printer builds an object **layer by layer**	**Proof:** explain it to a partner using a stack of paper
2	**Shapes + holes:** combine **solid shapes and holes** in Tinkercad to design a name badge	**Proof:** a badge with my name and a keyring hole
3	**Measure:** measure a real part in **millimetres** with a ruler	**Proof:** my sensor measurements written on the sheet
4	**Fit:** design a sensor mount that **fits** using measurements and a small gap	**Proof:** my mount file has 2 holes the right size and distance apart

Objective 1 · Layers · video

Video time!



How 3D Printers Work

Nat Geo Kids · about 3 minutes · youtu.be/HlvK6DLwCz4

Watch for...

1. What comes out of the printer's **nozzle**?
2. Does the printer build from the top or the **bottom**?
3. Name **one thing** people have 3D printed.

Scan to watch

After the video: Turn to your partner. Explain how the printer made the object in **one sentence** using the word **layer**.

Objective 1 · Layers

A 3D printer builds in layers

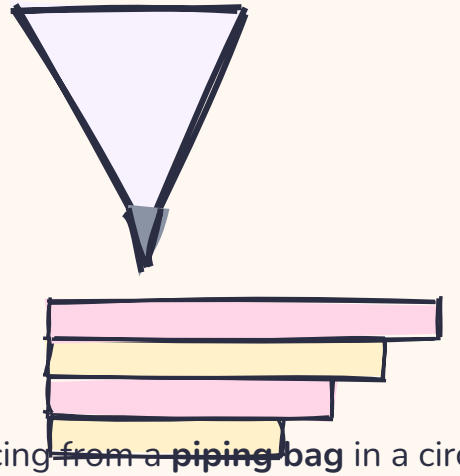
The idea

The printer melts a plastic string called **filament** and squeezes it out of a hot **nozzle**, like toothpaste. It draws one thin flat **layer**, moves up a tiny bit, and draws the next one on top.

Each layer is about **0.2 mm** thick. A 1 cm tall badge needs about **50 layers**!

Hot!
PLA plastic is made from corn starch. It melts at about 200 °C, so never touch the nozzle!

Example: decorating a cake



A baker squeezes icing ~~from a **piping bag**~~ in a circle, then another circle on top, then another. Slowly a tall swirl grows. The nozzle of a 3D printer does exactly the same with hot plastic.

Real world

Who 3D prints things?



Doctors
Custom hands and splints that fit one patient



Space engineers
Spare tools printed on the space station



Car makers
Test parts in hours instead of weeks



Makers like us
Robot parts, phone stands, cable clips



What would YOU print to fix a problem at home or at school?

Prototype
Engineers print a quick test version, called a prototype, check it, then improve it.

Objective 2 · Shapes + holes

Solid + hole = a new shape

The idea

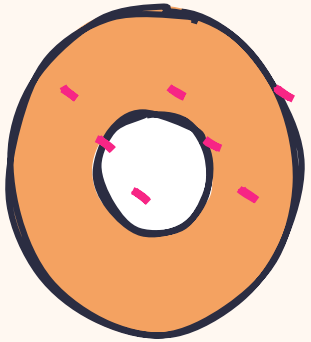
In Tinkercad every shape can be a **Solid** or a **Hole**.

Put a hole shape inside a solid, select both and press **Group** (Ctrl + G). The hole **cuts away** that part. Simple shapes + holes can make almost anything!

Look for stripes

Holes show as grey stripes. They only cut when you group them with a solid.

Example: a doughnut



solid

+ hole

A doughnut is just a big **solid cylinder** with a small **hole cylinder** in the middle. A keyring badge is a **solid box** with a **hole cylinder** near one end.

Objective 2 · Shapes + holes · Tinkercad tour

Meet Tinkercad (free, in the browser)

Do this	To…
Drag a shape	from the right panel onto the blue workplane
White handles	change length and width
Black handle (top)	changes height
Black cone	lifts the shape off the workplane
Curved arrows	rotate
View cube (top left)	look from the top, front or side
Ruler tool	type exact sizes in mm

Open it

Go to **tinkercad.com**, log in with the class code, then **Create → 3D Design**.

Snap grid

Set the snap grid (bottom right) to **1 mm** so sizes come out exact.

Lost the view?
Press F to zoom to your shape. Right-drag to spin around it.

Objective 3 · Measure

Engineers measure in millimetres

The idea

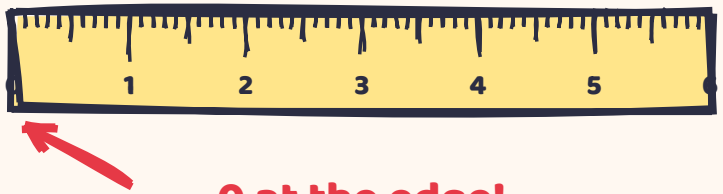
1 cm = 10 mm. Engineers use **mm** because parts must be exact.

How to measure well:

1. Put the **0** of the ruler at the edge, not the end of the ruler.
2. Look **straight down**, not from the side.
3. Measure **twice**. If you get two answers, measure a third time.

Units!
Write every number with its unit: 45 mm, not just 45.

Example: a phone case



0 at the edge!

A phone case that is **2 mm too small** will not go on. One that is 2 mm too big will fall off. The designer measured the phone very carefully first.

LEARN

Objective 4 · Fit

Fit needs a small gap

The idea

If a hole is **exactly** 16 mm and the sensor eye is 16 mm, it will be **too tight**: printers are never perfect.

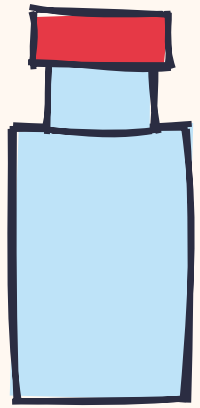
Engineers add a small gap called **tolerance**. For our printer: make holes **0.5 mm bigger**.

16 mm eye → **16.5 mm hole**.

Everywhere!

Tolerance is why your pencil fits the sharpener and your key fits the lock.

Example: a bottle lid



lid a little bigger = fits!

A bottle lid is made a tiny bit bigger than the bottle neck. Too small: it will not screw on. Too big: it leaks. The right small gap makes it fit perfectly.

Quick quiz

Match the word to its meaning

Word	Meaning
1. Filament	A. A part cut out of a solid shape
2. Layer	B. The small extra gap that makes parts fit
3. Hole	C. The plastic string the printer melts
4. Tolerance	D. A quick test version of a design
5. Prototype	E. One thin flat slice of the print

Answers

Cover this! 1C · 2E · 3A · 4B · 5D

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Slice detective
2. Make a name badge
3. Make it yours!
4. Measure the distance sensor
5. Design the sensor mount
6. Get it ready to print
7. Does it fit?

Class game

Slice detective

How to play

1. The teacher shows the **layers** of a secret object, one slice at a time.
2. Slice 1: a big circle. Slice 2: a smaller circle…
3. Teams whisper a guess after each slice.
4. Fewest slices to guess right wins a point!
5. Then YOU draw 4 slices of an object for another team.

Then discuss

Which objects are easy to guess from slices? Which are hard? Why?



Big idea

Every 3D shape can be cut into flat layers. That is how the printer 'thinks' about your design.

MAKE

Objective 2 · Build step 1 of 4

Make a name badge

a Drag a **Box**. Make it **60 mm** long, **25 mm** wide and **3 mm** tall.

b Drag **Text**. Type your name. Make it **3 mm** tall and lift it to sit **on top** of the box (height 3 mm).

c Drag a **Cylinder Hole**, 5 mm wide. Put it near one end for a keyring.

d Select everything, press **Group**. Choose a colour.

e Too long a name? Make the box longer or the letters smaller.



Check: Your badge is 60 × 25 × 3 mm, the name stands up 3 mm and there is a hole all the way through.

Text thinner than **2 mm** is too fragile to print.

Objective 2 · Shapes + holes · design time

Make it yours!



Add a symbol

A star, heart or robot from the Shapes panel next to your name



Two colours

Make the text a different colour from the base (it prints in one colour, but looks great on screen)



Change the outline

Round the corners with a Rounded Box, or make a circle badge

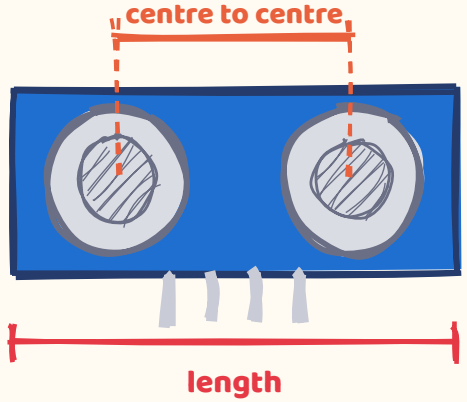


Gallery walk: leave your design on screen. Walk around and give one **star** (something great) and one **wish** (something to improve) to two other designers.

Objective 3 · Measure · your turn

Measure the distance sensor

What to measure	Typical	Our team
Board length	about 45 mm	_____ mm
Board height	about 20 mm	_____ mm
Each 'eye' (across)	about 16 mm	_____ mm
Centre of eye to centre of eye	about 26 mm	_____ mm
How far the eyes stick out	about 12 mm	_____ mm



Team jobs: Measurer holds the ruler · Checker measures again · Recorder writes it down. Swap jobs after each row!

Centre to centre?

Measure from the **left edge** of one eye to the **left edge** of the other. That equals centre to centre!

Objective 4 · Build step 2 of 4

Design the sensor mount

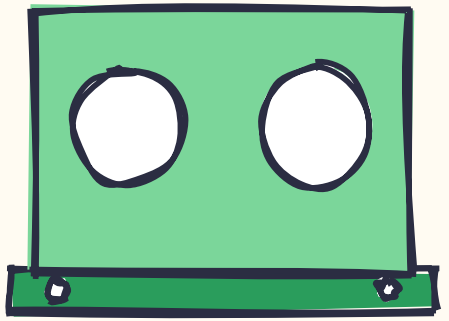
a **Back plate:** Box 50 mm long, 3 mm thick, 25 mm tall. This holds the sensor.

b **Eye holes:** two Cylinder Holes, **16.5 mm** wide, placed **26 mm apart** (centre to centre). Use YOUR measurements!

c **Base:** Box 50 × 20 × 3 mm lying flat, joined to the bottom of the back plate like an **L**.

d **Screw holes:** two 3.5 mm Cylinder Holes in the base to bolt it to the robot.

e Group everything. Check every size with the **Ruler** tool.



Check: Two eye holes, 16.5 mm wide, 26 mm apart, and two screw holes in the base.

Holes must go **all the way through**: make them taller than the plate.

MAKE

Objective 1 · Build step 3 of 4

Get it ready to print

a Click **Export → .STL**. Name it **JM10_team_mount**.

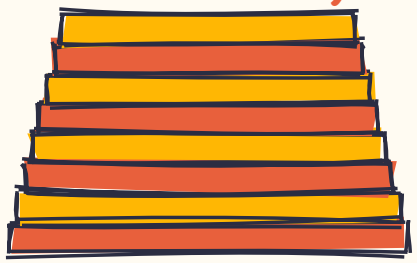
b The teacher opens it in **UltiMaker Cura**, the slicer. It cuts the design into layers!

c Lay the part **flat side down** so it does not need supports.

d Cura says how long it takes: a badge is about **20 minutes**, a mount about **40 minutes**.

e Best designs from each class are printed and fitted on the JM-11 robot.

slicer = cuts into layers



Check: The STL file is saved and the slicer shows the layers and the print time.

Test like an engineer

Does it fit?

Test	Expect	Result
Badge: name is readable	yes	pass / fail
Keyring goes through the hole	yes	
Sensor eyes push into the holes	snug, not loose	
Base sits flat on the robot	no wobble	
Screws line up with the robot holes	yes	

Debug checklist

Too tight? Make the holes 0.3 mm bigger.
Too loose? Make them 0.3 mm smaller.
Wrong distance apart? Measure again!

Real engineers **expect** the first print to be a bit wrong. Change one size, print again. That is called **iterating**.

Level up!

Designer challenges

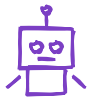
Choose your challenge



Phone stand: a stand that holds a phone at an angle for watching videos.



Cable clip: a clip that holds your USB cable on the desk edge.



Robot nameplate: a badge with your team name that bolts on the robot.



Pencil topper: a shape with a hole that fits YOUR pencil (measure it!).

Think like an engineer

Why do engineers design on a computer first instead of just making the part?

Boss level

Design a mount for TWO sensors at different heights.

Objective check

Can I really do it? Prove it.

01	Explain to a partner how a 3D printer makes an object. Use the word **layer**.	○ ○ ○
02	Show your badge: is there a **hole** that goes all the way through?	○ ○ ○
03	Read out one measurement of the sensor **with its unit**.	○ ○ ○
04	Point to the **tolerance** in your mount: which number did you make bigger, and why?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Ask 3 students to swap and check a partner's mount sizes with the Ruler tool.

Show what you know

What did we learn?

- [x] A 3D printer builds an object layer by layer
- [x] Solid shapes + holes can make almost anything
- [x] Engineers measure in millimetres, twice
- [x] Holes need a small gap (tolerance) to fit
- [x] First prints are prototypes: test, then improve

Exit ticket: why did we make the eye holes 16.5 mm and not 16 mm?

At home
Find one object at home that has a hole designed to fit something. Measure both!

YOU!

Next class: build a Line Following Robot, with your mount on it!